

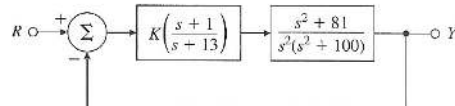
MAE 143B Linear Control HW7

and determine the value of the root-locus gain for which the complex conjugate poles have a damping ratio of 0.5.

5.13 For the system in Fig. 5.54,

- Find the locus of closed-loop roots with respect to K .
- Is there a value of K that will cause all roots to have a damping ratio greater than 0.5?
- Find the values of K that yield closed-loop poles with the damping ratio $\zeta = 0.707$.
- Use MATLAB to plot the response of the resulting design to a reference step.

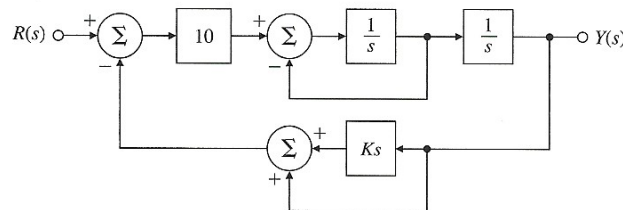
Figure 5.54
Feedback system for
Problem 5.13



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5.14 For the feedback system shown in Fig. 5.55, find the value of the gain K that results in dominant closed-loop poles with a damping ratio $\zeta = 0.5$.

Figure 5.55
Feedback system for
Problem 5.14



(b) Use MATLAB to plot the system's response to a reference step.

5.31 Consider the rocket-positioning system shown in Fig. 5.62.

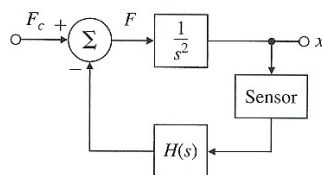
- Show that if the sensor that measures x has a unity transfer function, the lead compensator

$$H(s) = K \frac{s+2}{s+4}$$

stabilizes the system.

- Assume that the sensor transfer function is modeled by a single pole with a 0.1 sec time constant and unity DC gain. Using the root-locus procedure, find a value for the gain K that will provide the maximum damping ratio.

Figure 5.62
Block diagram for
rocket-positioning
control system



5.32 For the system in Fig. 5.63